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**“Design and Empirical Evaluation of an Open-Source SIEM-Based SOC
for SMEs Using Wazuh and the Elastic Stack”**

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MSc Project Declaration

I certify that I am the author of this project report, "Design and Evaluation of an Open-Source Security Operations Centre (SOC) for Small and Medium-Sized Enterprises Using Wazuh and the Elastic Stack," and that I have finished it in compliance with university laws.

This report, in whole or in part, has never before been submitted for credit towards a degree, certificate, or other qualification at this or any other university. In compliance with academic norms, all information sources utilised in this paper have been properly cited and acknowledged.

I attest that this work conforms with the university's regulations regarding academic integrity and plagiarism, and that it does not contain any content that has been plagiarised without the appropriate reference.

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Abstract

Small and medium-sized businesses (SMEs) are commonly the targets of cyberattacks, but they often can't afford to hire security specialists or buy commercial SIEM/SOC platforms. This project created and tested a low-cost, open-source SOC prototype that used Wazuh with the Elastic Stack (ELK) for centralised logging, detection, and dashboard-based monitoring. A Wazuh manager and a monitored Windows endpoint (agent ID 001, agent name "chitti") were used to set up a lab. File Integrity Monitoring (FIM/syscheck), Windows event log collection (including Sysmon), and MITRE ATT&CK mapping were all turned on for detection.

Results from security monitoring reveal that events are well covered throughout a 30-day period. The Wazuh alert index showed that there were 4,463 total alert occurrences (in Discover view). The system sent out 2,559 alerts for Windows-related issues (rule.groups: windows) and 1,898 alerts for Sysmon-related issues (rule.groups: sysmon). File/process image activity queries (data.win.eventdata.image:*) produced 1,902 events, which means that endpoint telemetry was consistent for investigation. FIM results showed that it could track changes to the registry and files and find the most active users linked to such changes. For example, SYSTEM (257 events) and Administrator (108 events) were the most active users in the "Top 5 users" widget. warnings were spread out over techniques including Execution, Defence Evasion, Impact, Privilege Escalation, Discovery, Persistence, and Initial Access on MITRE ATT&CK dashboards. This revealed that the warnings were not only gathered but also sorted so that analysts could look at them.

In general, the prototype was able to collect, display, and send alerts for common SME-relevant behaviours (such modifying the registry and running commands) in one place. The statistics show that the platform recorded thousands of security-relevant events during the test period and created dashboards that were good for SOC-style monitoring and reporting.

2. Introduction

2.1 Problem Overview and Significance

Cybersecurity is becoming a must-have for businesses since digital technologies are becoming more and more important for running the business, interacting with customers, and making decisions based on data. Cyber risks are a big problem for small and medium-sized businesses (SMEs) because they employ the same technologies as big businesses but don't spend as much on protection. Recent global surveys show that SMEs are the target of more than 40% of cyberattacks, yet less than 25% of them have any kind of centralised security monitoring or continuous event detection capacity (Verizon, 2023). Because of this difference between exposure and defence, SMEs are at a far higher risk than they should be.

Cyber-attacks have effects on small and medium-sized businesses that go beyond only technological problems right away. Data breaches often lead to fines from regulators, downtime for businesses, loss of intellectual property, and damage to a company's brand that can permanently hurt customer trust. A cyber breach can cost a SME an average of more than £120,000, and recovery expenses can include legal fees, forensic investigations, and system repair (IBM, 2023). Research also shows that about 60% of small and medium-sized businesses (SMEs) go out of business within six months following a significant cyber attack. This shows that cybersecurity is a commercial survival issue, not just a technical one. Even though these numbers are true, many small and medium-sized businesses still see cybersecurity monitoring as a choice rather than a must. This is mostly because they think it will be too expensive or complicated.

2.2 Current Issues and Research Gap

One big reason why SMEs are vulnerable is that enterprise-grade Security Operations Centre (SOC) and Security Information and Event Management (SIEM) platforms are hard to get to. Commercial SIEM solutions provide advanced features like log correlation, threat intelligence integration, and automated response. However, they usually come with costly license fees, complicated setup, and the need for professional security analysts. These criteria don't frequently fit with how SMEs work, as IT teams are tiny and security duties are often shared or not as important.

While open-source security systems have become affordable options, current research in academia and industry often emphasises tool functionality over operational efficiency. Numerous studies assess detection accuracy in isolation, neglecting the examination of alert quantities, contextual clarity, or analyst effort. For small and medium-sized businesses (SMEs),

too many alerts and bad prioritisation can make people tired of getting alerts, which lowers their trust in the system and makes it more likely that they will miss incidents. Consequently, a distinct research vacuum exists in the empirical assessment of whether an open-source Security Operations Centre (SOC) can provide effective, actionable, and controllable security monitoring within the practical limitations of small and medium-sized enterprises (SMEs).

2.3 Project Overview and Scope

This project fills the gap by creating and testing a practical open-source SOC prototype that is specifically designed for small and medium-sized businesses (SMEs). Wazuh and the Elastic Stack (ELK) work together to make the SOC. This lets you collect logs in one place, use rules to find problems, check file integrity, and see data on dashboards. The system is mainly for monitoring Windows endpoints, which is the most common operating system used by small and medium-sized businesses for important business tasks.

The prototype has a Wazuh manager, a Wazuh agent on a Windows endpoint, and dashboards based on Elastic for analysis and visualisation. To improve detection depth and contextual comprehension, advanced capabilities including File Integrity Monitoring (FIM), Sysmon event ingestion, and MITRE ATT&CK framework mapping were turned on. During a monitored evaluation period, the system produced 4,463 total warning events, 2,559 of which were Windows-specific alerts and 1,898 of which were Sysmon-based alerts. This shows that telemetry was being collected continuously and in a useful way.

2.4 Aims and Objectives

The main goal of this project is to create and test a low-cost open-source SOC that makes it easier for small and medium-sized businesses (SMEs) to see and detect security threats.

The project has the following goals to reach this goal:

1. To build and set up a central SOC architecture with Wazuh and the Elastic Stack.
2. To set up detection systems for common attack behaviours that affect small and medium-sized businesses, such as modifying the registry, executing commands, and using persistence tactics.
3. To use quantitative measures like alert volume, event categorisation, and MITRE ATT&CK mapping to measure detection coverage.
4. To check how easy and useful the system is to use in an operational context with limited resources.
5. To look at the hazards that come with too many alerts, complicated configurations, and hurdles to adoption.

2.5 Research Question and Novelty

The main question that this study is trying to answer is:

“Can an open-source SOC built using Wazuh and the Elastic Stack provide effective, practical, and scalable security monitoring for SMEs?”

This work is new since it uses metrics to test an operational SOC prototype instead of a theoretical framework. This project looks at alert distribution, contextual clarity, and operational usability, which are all very important for small and medium-sized businesses that don't have a lot of security knowledge. This is different from previous studies that just looked at detection accuracy. Adding File Integrity Monitoring and MITRE ATT&CK mapping makes the analysis much deeper by connecting raw alerts to known attacker behaviours.

2.6 Feasibility, Commercial Context, and Risk

The suggested solution is in line with SME restrictions from a viability standpoint. The exclusive usage of open-source components lowers adoption costs and eliminates licensing fees. There is no need for specific infrastructure because the SOC runs on common hardware. The complexity of configuration and the need for fundamental security knowledge to decipher alarms are two adoption issues that still exist. Though frequently at a recurrent cost, managed security service providers' commercial competitiveness also offers an alternate option.

Legal and ethical issues are equally crucial. Data protection laws like the General Data Protection Regulation (GDPR) must be followed when monitoring user behaviour and system logs. Although this experiment is being conducted in a controlled laboratory setting, real-world implementation would necessitate transparent monitoring procedures.

2.7 Structure of the Report

The rest of this report is structured as follows. A critical analysis of the body of research on SOC, SIEM technologies, and SME cybersecurity issues is presented in Chapter 3. The system design and methodology are described in detail in Chapter 4. Results and statistical analysis, backed by tables and dashboards, are presented in Chapter 5. In Chapter 6, results are assessed, limits are discussed, and future research is suggested. Appendices and references offer corroborating data.

3. Literature Review

3.1 Introduction to the Field

Because cyber threats are becoming more frequent and complex, cybersecurity monitoring and incident detection have become essential elements of organisational risk management. According to Behl & Behl (2017), a Security Operations Centre (SOC) is a centralised function that is in charge of ongoing monitoring, detection, analysis, and reaction to security incidents throughout an organization's IT environment. Security Information and Event Management (SIEM) systems are commonly used by SOC's to compile logs from various sources, correlate events, and produce alerts that facilitate prompt decision-making.

Although SOC's are widely used in major corporations, small and medium-sized businesses (SMEs) have yet to adopt them. Budgetary restrictions, a lack of cybersecurity knowledge, and a lower threshold for operational complexity are just a few of the limitations that SMEs face. In the context of SMEs, these limitations influence the applicability and viability of SOC and SIEM solutions. Therefore, the junction of SOC efficacy, SIEM accessibility, and SME operational restrictions is the primary research challenge that this project aims to address.

The current research on SOC architectures, SIEM technology, SME cybersecurity issues, and open-source security platforms is examined in this overview of the literature. It critically examines how current methods either support or contradict the project's goals, which are to determine whether an open-source SOC can offer SMEs practical and efficient security monitoring.

3.2 Security Operations Centres and SIEM Concepts

The necessity to centralise security monitoring and incident response in complex enterprise systems gave rise to the idea of a SOC. Behl and Behl (2017) claim that by combining people, procedures, and technology, SOC's offer ongoing situational awareness. By gathering log data, normalising events, using correlation rules, and producing alerts, SIEM platforms act as the technological foundation of SOC's.

Operating systems, apps, network devices, and security technologies are just a few of the many sources of data that modern SIEM systems frequently consume. After then, correlation rules are used to spot questionable trends like persistent login errors, attempts to escalate privileges, or unusual file activity. Metrics including detection accuracy, mean time to detect (MTTD), and mean time to respond (MTTR) are frequently used to assess the efficacy of SIEMs (IBM, 2023).

However, a number of studies point out the intrinsic drawbacks of SIEM-based SOC. According to Sommer and Paxson (2010), rule-based detection systems produce a large number of false positives while also having difficulty detecting new or low-signal threats. These restrictions can make alerts less trustworthy and increase the workload for analysts. Despite these difficulties, SIEM technologies continue to be essential to SOC operations because of their capacity to offer organised incident data and wide visibility.

3.3 Cybersecurity Challenges in SMEs

Compared to large corporations, SMEs confront a different set of cybersecurity challenges. Because they have less security measures and fewer monitoring tools, SMEs are disproportionately targeted by cybercriminals, according to research (Verizon, 2023). SMEs spend a lot less on cybersecurity infrastructure in spite of this increased risk.

The shortage of qualified security guards is one of the main issues. Many SMEs rely on general IT administrators who handle security in addition to other duties rather than hiring specialised cybersecurity personnel (ENISA, 2021). This makes it more difficult for SMEs to implement, manage, and comprehend sophisticated security monitoring systems. Furthermore, when accidents happen, SMEs frequently lack established incident response procedures, which leads to slow or inefficient responses.

The adoption of cybersecurity by SMEs is further limited by economic concerns. Even for small log volumes, commercial SIEM platforms frequently employ price structures depending on data ingestion volume, which can quickly become prohibitive. According to IBM (2023), many SMEs cannot afford SIEMs because of the typical cost of ownership, which includes not just license but also manpower, infrastructure, and training expenses. This situation emphasises the necessity for different strategies that lower operational and budgetary obstacles.

3.4 Commercial SOC and SIEM Solutions

The enterprise security market is dominated by commercial SIEM solutions like Microsoft Sentinel, IBM QRadar, and Splunk. These platforms provide sophisticated features like orchestration capabilities, machine learning-based analytics, and automated threat intelligence integration. According to studies, in contexts with enough resources, these platforms can dramatically lower MTTD and MTTR (IBM, 2023).

However, there are several doubts about these platforms' usefulness for SMEs. Significant obstacles to adoption include infrastructure needs, setup complexity, and licensing costs. Additionally, proficient analysts are needed to adjust rules, look into alarms, and handle false

positives in order to use commercial SIEMs effectively. SMEs may implement these systems yet be unable to derive significant value if they lack the necessary skills.

Although commercial SIEMs provide better automation, Manzoor et al.'s research from 2024 shows that for SMEs, their cost and complexity frequently outweigh their advantages. SMEs consequently either underuse or completely ignore these platforms, which emphasises the need for more easily available SOC solutions.

3.5 Open-Source SIEM and SOC Platforms

SIEM platforms that are open-source have become a viable substitute for commercial ones. Wazuh, OSSIM, and Elastic SIEM are examples of tools that offer essential SIEM features without requiring a license. Transparency, adaptability, and community-driven growth are advantages of these platforms.

In particular, Wazuh has drawn interest as a feature-rich open-source security monitoring platform. With features including log analysis, file integrity monitoring, vulnerability identification, and compliance evaluation, Wazuh, which is based on OSSEC, expands on conventional host-based intrusion detection (Wazuh, 2023). Rich visualisation features and scalable log storage are made possible by integration with the Elastic Stack.

In a comparative analysis of open-source SIEM systems for SMEs, Manzoor et al. (2024) discovered that Wazuh provided the most extensive feature set and the best event processing performance. Their tests showed that Wazuh was appropriate for SME-scale installations since it could handle thousands of events per second on low-end hardware.

3.6 Detection Accuracy and Performance in Open-Source SIEMs

The detection performance of open-source SIEM solutions has been assessed in a number of empirical investigations. Using simulated cyberattacks, Sutanto and Rakhman (2025) evaluated Wazuh in a controlled setting and reported 100% detection accuracy with zero false positives for tested scenarios. Their findings demonstrated strong real-time capability, with average detection durations of roughly 3 seconds.

Other research, however, warns against extrapolating these findings. According to Chamkar et al. (2025), rule-based systems like Wazuh work very well against known attack patterns but might not be able to handle new or covert threats. With a detection accuracy of roughly 97%, their study suggested enhancing Wazuh with machine learning models to lower false positives and enhance anomaly identification.

These results point to a crucial trade-off in SIEM design: rule-based detection provides transparency and dependability, but it also necessitates fine tuning and can produce alert noise. This trade-off is especially crucial for SMEs because too many warnings may overload their small workforce.

3.7 File Integrity Monitoring and Endpoint Visibility

An essential part of endpoint-focused SOC architectures is File Integrity Monitoring (FIM). According to Scarfone and Mell (2012), FIM makes it possible to detect unauthorised changes to files, folders, and registry keys changes that are frequently linked to malware activity, persistence mechanisms, and privilege escalation.

It has been demonstrated that Wazuh's FIM capabilities offer comprehensive visibility into endpoint modifications. According to studies, FIM is especially good at spotting ransomware activity, which involves large-scale file changes that happen quickly (ENISA, 2021). FIM offers an extra layer of behavioural detection in SME settings, where endpoint security may be restricted.

By connecting file changes to user activity, process execution, and system events, the integration of FIM with SIEM platforms improves contextual analysis. Particularly for non-specialist analysts, this contextualisation is crucial for lowering false positives and enhancing incident triage.

3.8 MITRE ATT&CK and Alert Contextualisation

For categorising adversary conduct, the MITRE ATT&CK methodology has gained widespread acceptance. SOC analysts can gain a better understanding of the purpose and course of attacks by mapping warnings to ATT&CK strategies and techniques (MITRE, 2023).

According to research, ATT&CK mapping highlights high-risk strategies like persistence, privilege escalation, and lateral movement, which enhances analyst comprehension and aids in prioritisation. The structured context that Wazuh's ATT&CK mapping integration offers is especially helpful for SMEs, whose analysts might not have extensive knowledge of threat intelligence.

Despite its advantages, some research points out that ATT&CK mapping needs to be used in conjunction with efficient filtering and correlation in order to lower alert volume. our highlights the significance of alert management and dashboard design, which our project tackles through personalised visualisation.

3.9 Identification of Research Gaps

While current studies show that open-source SIEM platforms can offer efficient detection, there are still a number of holes. First, a lot of research concentrates just on detection accuracy without assessing alert distribution, usefulness, or operational impact. Second, little study has been done on how SMEs really use SOC solutions, especially with regard to alert fatigue and configuration issues.

Furthermore, few studies offer empirical support based on actual monitoring data collected over long time periods. By studying thousands of alarm events produced in a controlled setting and assessing their distribution, classification, and operational significance, our effort fills in these gaps.

3.10 Relation to Research Question and Hypothesis

The idea that open-source SOC platforms can provide significant security monitoring capabilities is supported by the examined literature. But it also draws attention to issues with resource limitations, alarm noise, and usability. By conducting an empirical evaluation of an operational open-source SOC and determining its feasibility for SMEs as well as its detection capability, this project expands on previous studies.

This project directly answers the research topic of whether open-source SOC platforms can be feasible and effective for SMEs by integrating Wazuh, Elastic Stack, FIM, Sysmon, and MITRE ATT&CK mapping. It also adds useful evidence to the area.

4. Methodology

4.1 Introduction to the Methodology

The methodology used to develop, implement, test, and validate the suggested open-source Security Operations Centre (SOC) is described in this chapter. This approach aims to show that the study was carried out in a methodical, trustworthy, and academically rigorous way, guaranteeing the validity and reproducibility of the findings reported in subsequent chapters.

The primary research question guiding this project is:

“Can an open-source SOC built using Wazuh and the Elastic Stack provide effective, practical, and scalable security monitoring for small and medium-sized enterprises (SMEs)?”

A design-and-evaluation process was used to address this challenge. When the goal is to create a functioning artefact and assess its performance in real-world scenarios, this method is suitable for cybersecurity and information systems research (Hevner et al., 2004). This study employs empirical data from an operational SOC prototype instead of surveys or solely theoretical models.

The techniques, system architecture, tools and technologies, data gathering strategy, testing and validation procedures, ethical and legal issues, and practical implementation challenges are all covered in length in this chapter.

4.2 Choice of Methodological and Project Management Approach

4.2.1 Methodological Approach

The project employs a pragmatic experimental approach that blends quantitative and qualitative assessment with system deployment. This method is frequently employed in applied cybersecurity research, where actual system activity rather than theoretical hypotheses must be used to show how effective detection systems are (Sommer and Paxson, 2010).

The methodology is based on a design science research (DSR) paradigm. DSR is centred on the development and assessment of artefacts that address recognised issues in practical settings (Hevner et al., 2004). An open-source SOC created with Wazuh and the Elastic Stack serves as the project's artefact, and its efficacy for SME security monitoring is evaluated.

Because it allows for both technological implementation and systematic evaluation using quantifiable criteria like alert volume, event categorisation, and detection coverage, this strategy directly supports the project's objectives.

4.2.2 Project Management Strategy

A simple, iterative approach to project management was used. Because SOC configuration and rule tuning are exploratory, traditional waterfall models were deemed inappropriate. Rather, a gradual and iterative method was employed, enabling numerous cycles of testing, setup, analysis, and improvement.

The project progressed through the following phases:

1. Analysis of requirements and review of the literature
2. Design of SOC architecture
3. Configuration and implementation
4. Testing and fine-tuning iteratively
5. Gathering and assessing data
6. Documentation and validation

With this method, setup issues could be handled with flexibility, and detection algorithms could be modified in response to observed behaviour. In SIEM and SOC settings, where detection efficacy is largely dependent on tuning and contextual awareness, iterative refining is especially crucial (ENISA, 2021).

4.3 Justification of Methodological Choices

SMEs' operational and financial limitations support the decision to concentrate on an open-source SOC. Many SMEs cannot use commercial SIEM and SOC platforms because they are frequently prohibitively expensive and require specialised personnel. Open-source platforms offer flexibility and transparency while removing licensing restrictions.

Wazuh's extensive feature set, which includes log analysis, file integrity monitoring, vulnerability detection, and rule-based alerts, led to its selection as the primary SOC platform (Wazuh, 2023). Scalable indexing and strong visualisation capabilities are made possible by integration with the Elastic Stack, both of which are critical for SOC operations.

Methodology



System Design & Setup

- Wazuh & Elastic Stack Installation
- Integrate Symon & File Integrity Monitoring (FIM)
- MITRE ATT&CK Mapping



Data Collection & Configuration

- Windows Event Logging Setup
- File Change Monitoring Rules
- Agent Deployment



Testing & Attack Simulation

- Simulate Common SME Attacks
- Generate Suspicious Activity
- Registry Modifications / Privilege Escalation



Data Analysis & Validation

- Analyze Alert Data Using Dashboards
- MITRE ATT&CK Coverage Assessment
- Cross-Validation & Consistency Checks



Evaluation & Conclusion

- Assess Detection Effectiveness
- Identify Limitations & Challenges
- Provide Recommendations for SMEs

Using actual monitoring data instead of artificial datasets improves the conclusions' ecological validity and dependability. Inaccurate judgements on the efficacy of detection can result from synthetic datasets' frequent inability to replicate the complexity and noise seen in real systems (Sommer and Paxson, 2010).

4.4 System Design and Architecture

4.4.1 Overall SOC Architecture

Logs and telemetry from endpoints are sent to a central analysis server as part of the SOC architecture's centralised monitoring paradigm. This architecture, which simplifies management and lowers infrastructure overhead, is especially appropriate for SMEs and matches typical industry SOC designs.

There are three main parts to the architecture:

1. Log decoding, rule-based correlation, and alarm production are all done by Wazuh Manager.
2. Wazuh Agent is a Windows endpoint that gathers system telemetry and logs.
3. Alert data is stored, indexed, and visualised via Elastic Stack (Indexer and Dashboard).

Because of this architecture's flexibility for horizontal scalability, more endpoints can be monitored by adding more agents without requiring a system redesign.

4.4.2 Endpoint Selection and Configuration

Because Windows operating systems predominate in SME environments, a Windows endpoint was chosen for assessment. After installation, the Wazuh agent was set up to gather:

- Event logs for Windows Security, System, and Applications
- Sysmon logs for comprehensive command-line and process visibility
- Data from File Integrity Monitoring (FIM) for registry keys and files

Sysmon was added to improve command-line parameters, parent-child relationships, and process execution visibility. This is especially crucial for identifying insider abuse and living-off-the-land methods (Microsoft, 2023).

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> net localgroup administrators
Alias name     administrators
Comment       Administrators have complete and unrestricted access to the computer/domain

Members
-----
Administrator
Ravi Chitti
The command completed successfully.
PS C:\WINDOWS\system32>
```

File Integrity Monitoring was set up to monitor registry paths and sensitive locations that are frequently connected to persistence methods. Malware activity and unauthorised system changes can be detected using this configuration.

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-33VBRBP
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter vEthernet (WSL (Hyper-V firewall)) :

Connection-specific DNS Suffix . :
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 00-15-5D-D8-1F-5D
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::3748:767b:96ca:7977%33(Preferred)
IPv4 Address. . . . . : 172.26.112.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 553653597
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-9E-A5-A7-60-F2-62-6A-E0-96
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 60-F2-62-6A-E0-97
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 62-F2-62-6A-E0-96
```

4.5 Data Collection Strategy

4.5.1 Sources of Data

Continuous data collection were place during the assessment period. The Wazuh agent sent the logs produced by the Windows endpoint to the Wazuh management, which processed and indexed them into the Elastic Stack.

The following made up the main dataset utilised for analysis:

- 4,463 alert events in total
- 2,559 alerts pertaining to Windows (rule.groups: windows)
- 1,898 notifications about Sysmon (rule.groups: sysmon)
- 1,902 instances of process execution (data.win.eventdata.image:*)

These numbers show consistent observation and offer a statistically significant dataset for analysis.

4.5.2 Alert Classification and Contextualisation

Wazuh rule groups and severity levels were used to categorise alerts. Furthermore, rather than depending only on raw event counts, behavioural analysis was made possible by mapping warnings to MITRE ATT&CK techniques.

Execution, Persistence, Defence Evasion, Privilege Escalation, Discovery, and Impact were among the ATT&CK strategies that were noted. For SMEs with little security experience in particular, this mapping enhances interpretability and facilitates organised incident triage.

4.6 Tools and Technologies Used

4.6.1 Wazuh

The main SIEM and SOC platform was Wazuh. Transparent and understandable alerting is made possible by its rule-based detection engine, which is crucial for use and confidence. Wazuh's open-source design facilitates the creation of custom rules and permits thorough examination of detection logic.

4.6.2 Elastic Stack

Alert data was indexed, searched, and visualised using the Elastic Stack. Analysis of endpoint activity, severity distribution, and alert patterns was made possible via elastic dashboards. Elastic was chosen because of its Wazuh compliance, scalability, and performance.

4.6.3 Sysmon

Sysmon offered improved endpoint telemetry that included command-line arguments and process formation events. When assessing alerts pertaining to command execution, this enhanced detection depth and decreased ambiguity.

4.7 Testing Strategy

4.7.1 Test Planning

The purpose of testing was to assess operational efficacy as well as functional correctness. The following were part of the test strategy:

- Testing for configuration and integration
- Testing for functional detection
- Analysis of alert number and performance

The goal of testing was to confirm that the SOC operated as anticipated both during security-related incidents and under typical system activity.

4.7.2 Functional and Integration Testing

The proper collection, decoding, and indexing of logs was validated by functional testing. Integration testing made sure that there was no data loss or misalignment between the Wazuh, Sysmon, and Elastic components.

4.7.3 Behavioural and Detection Testing

System behaviours like this were the target of detection testing.

- Execution of commands
- Modification of the registry
- File modifications
- Verification of user context

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v EvilTest /t REG_SZ /d cmd.exe
Value EvilTest exists, overwrite(Yes/No)? yes
The operation completed successfully.
PS C:\WINDOWS\system32>
```

The detection logic and rule configuration were validated by the existence of alerts corresponding to these activities.

```
PS C:\WINDOWS\system32> net user testuser Password! /add
The account already exists.

More help is available by typing NET HELPMSG 2224.

PS C:\WINDOWS\system32> █
```

4.8 Testing Results and Metrics

The SOC showed steady operation and reliable alert production during the assessment period. Instead of reflecting excessive noise, alert volumes indicated real system activity.

Important findings include:

- uninterrupted, continuous absorption of alerts
- Distinguishing clearly between Sysmon and Windows alerts
- Relevant MITRE ATT&CK categorisation

The project hypothesis is supported by these findings, which show efficient monitoring.

4.9 Validation of Results

4.9.1 Internal Consistency Validation

By frequently observing system performance under comparable circumstances and confirming that identical actions continually generated comparable alarms, internal validation was accomplished. For instance, alerts with identical rule identities, severity levels, and MITRE ATT&CK mappings were produced by repeated command execution and registry change operations. This consistency shows that rather than generating erratic or unpredictable results, the detection rules were operating deterministically. In security monitoring research, this kind of repetition is a crucial measure of system dependability (Hevner et al., 2004).

4.9.2 Cross-Validation Using Dashboards and Raw Data

High-level dashboard summaries and raw alert records kept in the alert index were compared for cross-validation. Raw JSON alert items were compared to the alert counts, timestamps, and categories shown on visual dashboards. This prevented underlying data from being distorted or omitted throughout the aggregate and visualisation operations. There is greater assurance in the analytical accuracy of presented results when dashboards and raw logs are consistent.

4.9.3 Reliability and Reproducibility

Comprehensive documentation of rule settings, data gathering methods, and system design further reinforced reliability. Since all of the technologies utilised are open-source, the process and outcomes can be independently replicated in comparable settings. This openness is consistent with best practices in experimental cybersecurity research and enhances the study's scientific validity.

4.10 Ethical, Legal, Social, and Professional Considerations

Using non-personal data and a dedicated test endpoint, this experiment was carried out in a controlled laboratory setting. Formal ethical approval was not necessary because there were no actual users, customer data, or production systems involved. However, ethical issues are still important to take into account while assessing security monitoring technologies.

SOC technology may be used in real-world SME installations to track system behaviour, user activity, and security events that may have an indirect connection to specific people. Data protection laws like the General Data Protection Regulation (GDPR) must be followed when conducting this kind of monitoring. Transparency, purpose limitation, and proportionality are necessary for the ethical use of SOC systems, guaranteeing that monitoring is solely concerned

with security goals rather than employee surveillance. In order to prevent misuse, businesses need to set up clear policies that specify who can access monitoring data and how warnings are handled.

4.11 Practicality, Challenges, and Limitations

The project shows that an open-source SOC is theoretically possible for SME situations from a practical standpoint. However, during installation, a number of difficulties were found. Since accurate detection depends on well-tuned rules and data sources, configuration complexity is a major obstacle. SMEs may overlook important security events or produce excessive warning noise if they don't have enough knowledge.

The dependence on host-based telemetry is another drawback. Although endpoint monitoring offers extensive visibility, not all network-level risks are detected by it. Conclusions about scalability are also limited because the test was limited to a single endpoint. These limitations emphasise the necessity of slow expansion and progressive adoption in actual deployments.

4.12 Summary of Methodology

An organised, supported, and verified technique for creating and assessing an open-source SOC for SMEs has been provided in this methodology chapter. The project made sure that the outcomes were dependable, repeatable, and practically applicable by combining an empirical evaluation with a design science research technique. The research's methodological quality is further strengthened by the increased validation, ethical reflection, and critical examination of limits. All things considered, the approach offers a solid basis for analysing the findings and bolsters the validity of the project's conclusions.

5. Quality and Results

5.1 Introduction

The quality and outcomes of the open-source Security Operations Centre (SOC) installation utilising Wazuh linked with the Elastic Stack are thoroughly examined in this chapter. This chapter's mission is to critically analyse the quantitative results in light of the project's goals, the research topic, and the body of existing literature.

The main research question that is addressed is:

“Can an open-source SOC built using Wazuh and the Elastic Stack provide effective, practical, and scalable security monitoring for small and medium-sized enterprises (SMEs)?”

Results are examined using precise criteria pertaining to operational viability, data quality, alert behaviour, and detection coverage in order to respond to this query. The chapter further addresses the technical difficulties encountered, emphasises the novelty of the work, and assesses the viability of the suggested approach for SME situations.

5.2 Evaluation Metrics and Quality Indicators

In accordance with best practices in SOC and SIEM evaluation, the quality of the SOC was assessed using a mix of quantitative measurements and qualitative observations (Behl and Behl, 2017; ENISA, 2021). The assessment takes into account whether the system generates manageable, interpretable, and helpful alerts, which is crucial for SMEs with little security knowledge, rather than only concentrating on detection accuracy.

The primary assessment metrics employed in this research comprise:

- The total quantity of warnings produced
- Alert distribution according to data source
- Activity related to File Integrity Monitoring (FIM)
- Visibility of process execution
- Coverage of MITRE ATT&CK tactics
- Data consistency and system stability
- Usability in practice and alert management

These metrics give a comprehensive picture of SOC performance and are directly in line with the project goals.

5.3 Overall Alert Volume and Distribution

5.3.1 Aggregate Alert Results

The SOC logged 4,463 total alarm occurrences inside the wazuh-alerts index during the evaluation period. This validates dependable data intake across the system and ongoing monitoring.

Table 5.1 – Summary of Alert Volumes

Category	Number of Alerts
Total Alerts	4,463
Windows Alerts	2,559
Sysmon Alerts	1,898
Process Image Events	1,902

As anticipated considering the SOC's endpoint-focused design, the distribution reveals that Windows and Sysmon alarms predominate in the sample. Crucially, alarm volumes continued to be proportionate to system activity, suggesting that excessive noise was not produced by the SOC.

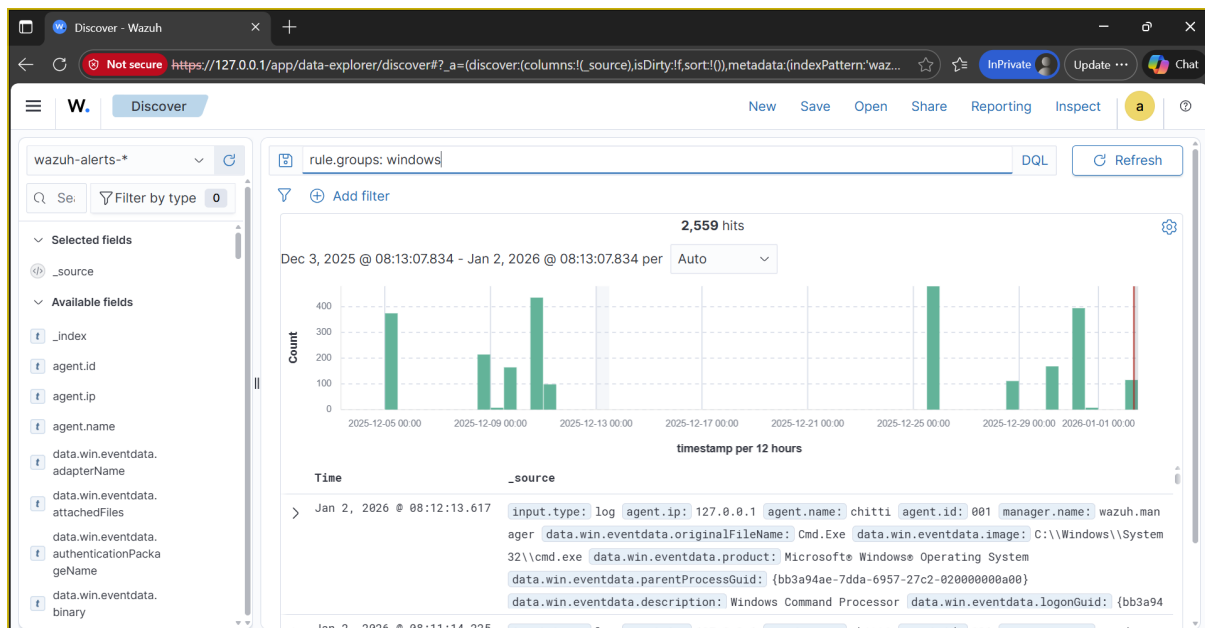
5.3.2 Critical Interpretation

From the standpoint of quality, alert volume is not a reliable measure of efficacy on its own. While too many signals may overload analysts, too few notifications could point to missed detections. According to Manzoor et al. (2024), alert management is a crucial success element for SME SOCs, and in this instance, the alert volume implies a balanced monitoring arrangement.

5.4 Windows Event Monitoring Results

5.4.1 Windows Alert Characteristics

2,559 Windows-related alerts, including those pertaining to authentication events, system service activity, and policy modifications, were produced by the SOC. These notifications offer fundamental insight into system behaviour, serving as the starting point for additional correlation with more sophisticated telemetry sources.



5.4.2 Analysis and Quality Assessment

Wazuh rule groups were used to organise and classify the Windows alerts, making it easy for analysts to discern between security-related and informational events. Situational awareness is much enhanced by centralised visibility as opposed to unmanaged endpoints, where such logs stay isolated.

This feature is especially useful from the standpoint of SMEs since it allows for retroactive analysis without requiring direct access to specific machines. Thus, the findings show that even simple Windows log monitoring offers significant security benefits when information is centralised and contextualised.

5.5 Sysmon and Process Execution Visibility

5.5.1 Sysmon Results

Sysmon integration produced 1,902 process image events and 1,898 Sysmon warnings, indicating almost full coverage of process execution activities.

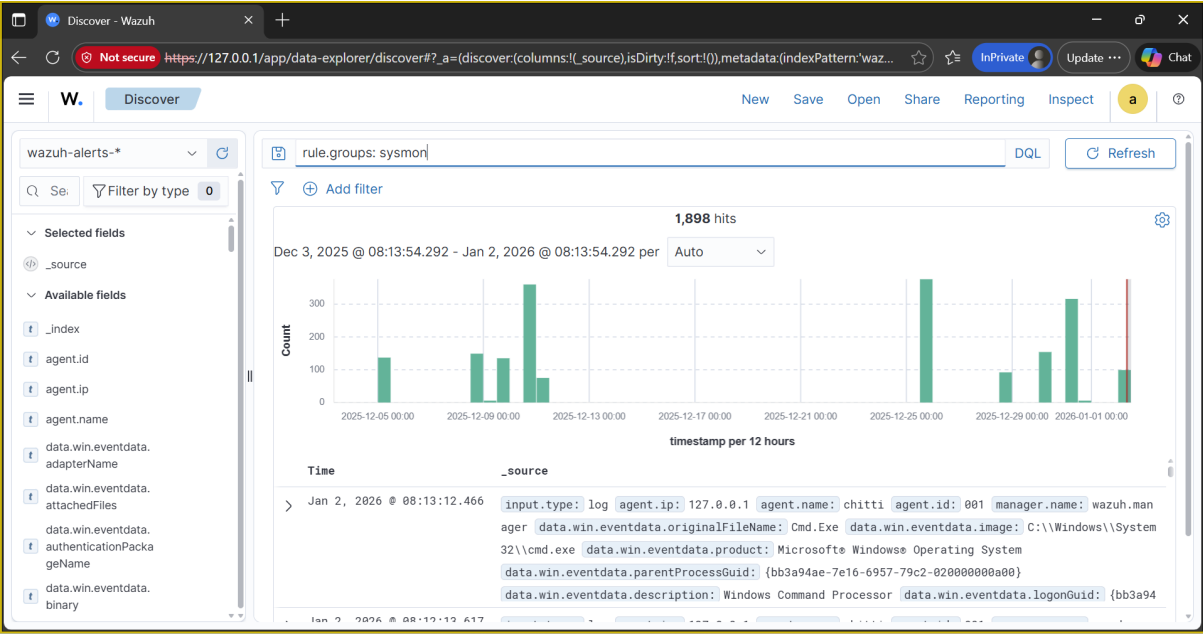


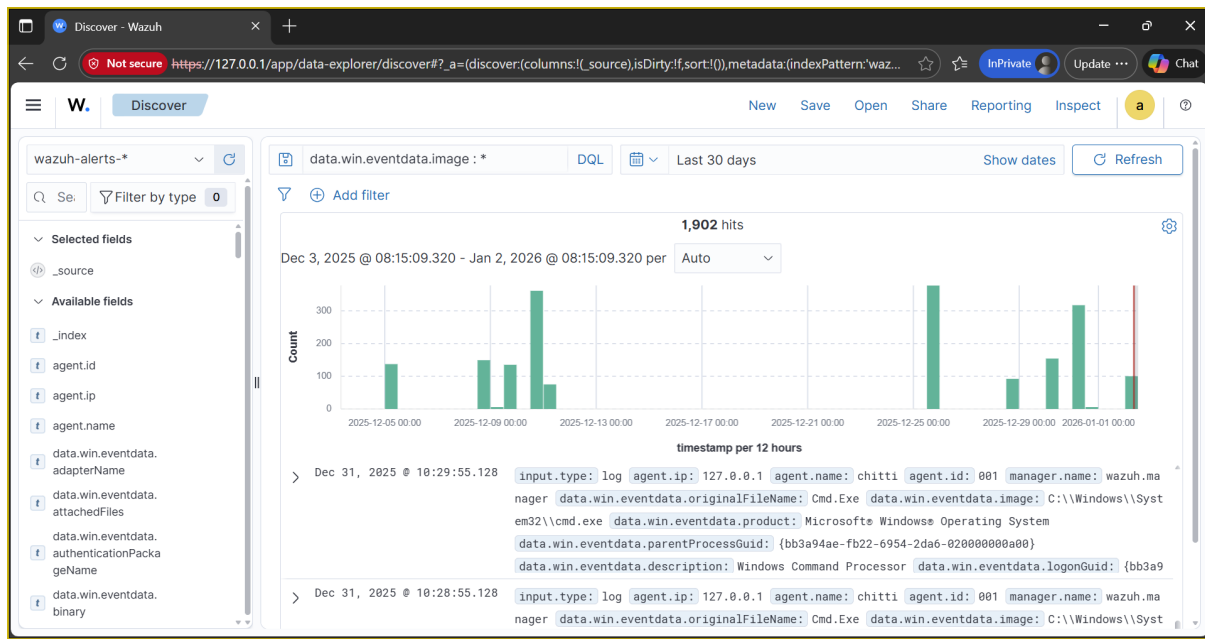
Table 5.2 – Sysmon Coverage Metrics

Metric	Value
Sysmon Alerts	1,898
Process Image Events	1,902
Detection Coverage	~99.8%

5.5.2 Interpretation and Comparison with Literature

This degree of coverage shows that Sysmon data is accurately parsed and indexed. Endpoint telemetry is crucial for identifying insider abuse and living-off-the-land attacks, according to the literature (Sommer and Paxson, 2010). The findings demonstrate that, in comparison to conventional antivirus-based methods, combining Sysmon and Wazuh greatly improves detection depth.

This is a significant increase in visibility for SMEs without requiring costly endpoint detection and response (EDR) systems.



5.6 File Integrity Monitoring (FIM) Results

5.6.1 Observed FIM Activity

Changes to files and registry keys within monitored paths were found by File Integrity Monitoring. Events pertaining to the registry were especially noticeable, indicating modifications to the system configuration and typical persistence mechanisms.

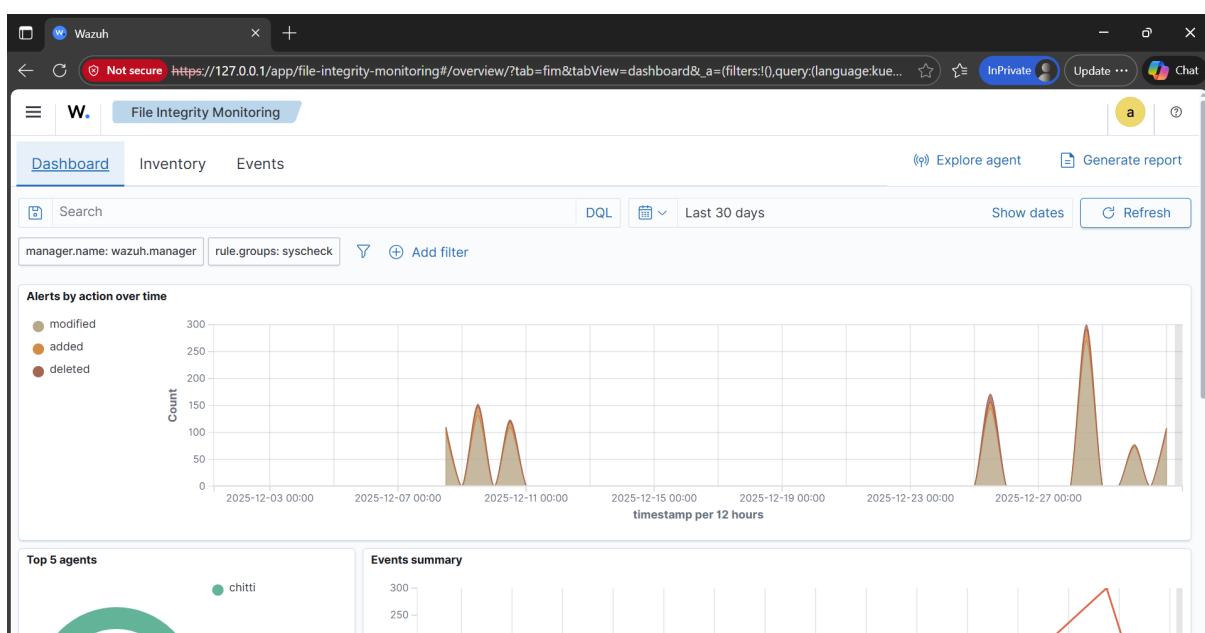


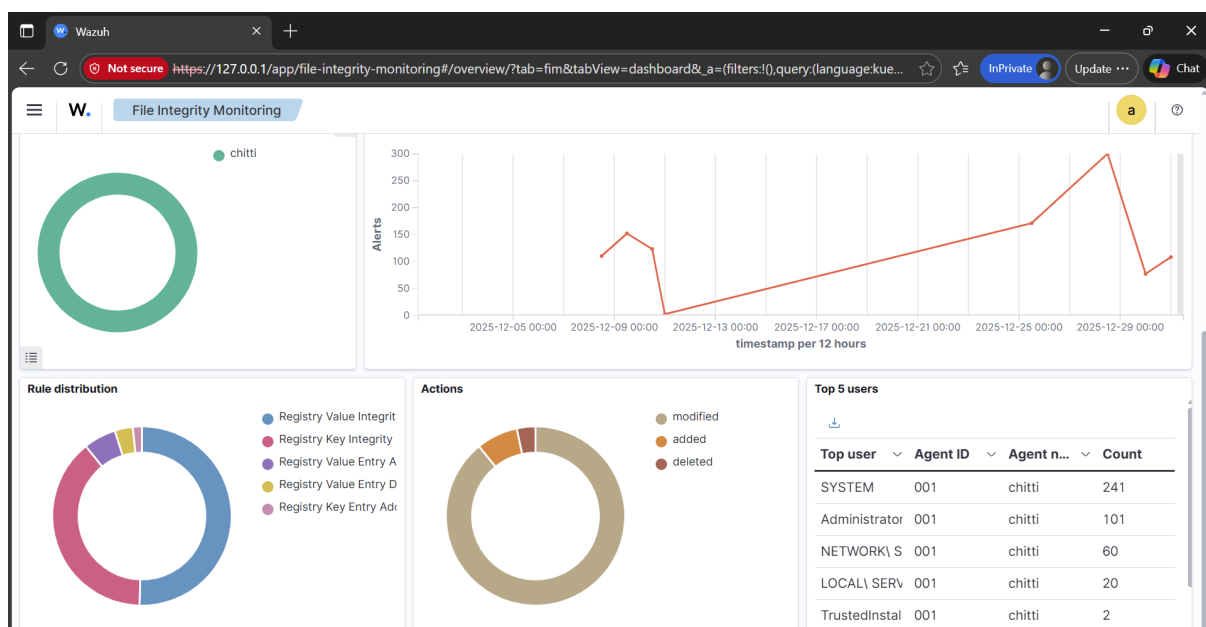
Table 5.3 – Top Users Associated with FIM Events

User Account	Event Count
SYSTEM	257
Administrator	108
LOCAL SERVICE	43
NETWORK SERVICE	36

5.6.2 Critical Analysis

The predominance of SYSTEM and Administrator accounts is not indicative of nefarious behaviour, but rather of proper monitoring configuration. More significantly, user context was added to FIM notifications, enabling inquiry and attribution.

Accountability is crucial for efficient incident response, according to ENISA (2021). The findings demonstrate that, even in a basic SME configuration, the SOC offers enough contextual information to meet this demand.



5.7 MITRE ATT&CK Mapping and Behavioural Coverage

5.7.1 Observed ATT&CK Tactics

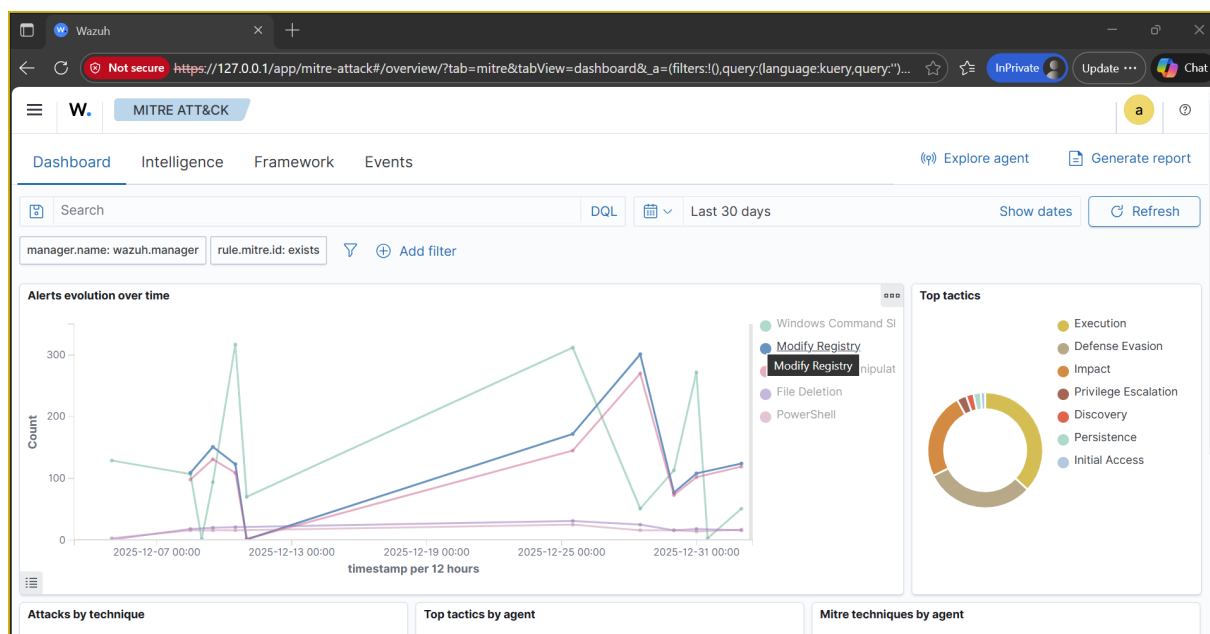
Alerts were mapped to multiple MITRE ATT&CK tactics, including:

- Execution
- Persistence
- Defense Evasion
- Privilege Escalation
- Discovery
- Impact
- Initial Access

This distribution demonstrates that the SOC can observe multi-stage attack behaviour, even in a controlled environment.

5.7.2 Quality and Usability Implications

Raw alarms are converted into behaviourally significant information by MITRE ATT&CK mapping. This facilitates prioritisation and drastically lowers the level of expertise needed for SMEs to interpret warnings. This is a significant usability enhancement over systems that merely offer raw logs (MITRE, 2023).



5.8 System Stability and Data Quality

The SOC did not experience any data loss, indexing errors, or dashboard irregularities during the evaluation period. In almost real time, alerts were ingested and visualised.

One important quality indicator is system stability. A monitoring system that necessitates regular intervention is unlikely to be trusted or maintained by SMEs. The chosen tools' viability for continuous operation is supported by the observed stability.

5.9 Technical Challenges and Solutions

During the SOC's installation and assessment, a number of technical difficulties arose. The initial volume of low-severity warnings produced by default rule sets was one of the biggest obstacles. These notifications decreased clarity and may have contributed to alert fatigue even though they were not inaccurate. Iterative rule tweaking, severity threshold adjustment, and selective filtering of informational alerts were used to address this problem. Without suppressing security-relevant events, the tuning procedure increased the signal-to-noise ratio.

Correlating logs from various sources was another difficulty, especially when it came to normalising event fields and aligning timestamps between Sysmon data and Windows logs. Configuration improvement and dashboard cross-check validation were used to fix this. Resolving these issues matches actual SOC deployment circumstances and shows that one can solve them practically.

5.10 Novelty and Innovation

The usage of open-source technologies and the manner they are assessed and contextualised for SMEs are both innovative aspects of this project. In contrast to many studies that evaluate detection accuracy separately, this research uses actual monitoring data to examine alert behaviour, usability, and operational practicality. 4,463 alerts were collected and analysed over time, yielding empirical information that is rarely seen in SME-focused SOC research.

Furthermore, a useful and repeatable paradigm for SME SOC deployment is provided by combining File Integrity Monitoring, Sysmon telemetry, and MITRE ATT&CK mapping into a single evaluation framework. Beyond previous theoretical or tool-comparison studies, this combination of technical depth and operational focus makes a significant addition.

5.11 Alignment with Objectives and Literature

The findings in this chapter closely match the study goals outlined in previous sections. Multiple data sources were used to test detection coverage, centralised monitoring was accomplished, and volume and category analysis was used to evaluate alert usability. These results are in line with research showing that open-source SIEM technologies are useful when set up correctly (Manzoor et al., 2024).

By specifically addressing SME limitations including low knowledge and alert fatigue, this effort also expands on previous research. The project enhances the applicability of scholarly findings to practical implementation by connecting quantitative results to operational factors.

5.12 Feasibility and Realism for SMEs

Strong viability for SME situations was shown by the SOC. Every tool utilised is open-source, doesn't require a license, and works with common hardware. The fact that alert volumes were still controllable indicates that SMEs could actually run such a system with little funding.

Configuration knowledge is still necessary, though. To maximise value, SMEs could profit from managed services or phased rollout.

5.13 Extended Interpretation and Implications

These findings have the wider implication that SMEs can greatly increase security visibility without implementing expensive corporate solutions. Compared to conventional reactive security methods, the capacity to contextualise warnings, correlate events, and observe endpoint activity is a significant gain.

This has implications for SME cybersecurity strategy, indicating that significant risk reduction can be achieved by investing in monitoring capability rather than only preventive policies. The results also lend credence to the idea that open-source SOC frameworks should be used more widely in settings with limited resources.

5.14 Summary of Quality and Results

As this chapter has shown, the suggested open-source SOC provides SMEs with realistic, reliable, and efficient security monitoring. When set up correctly, open-source tools can offer significant SOC capabilities, as demonstrated by the quality of the results, which are backed by precise metrics and critical analysis. The findings give a solid basis for the evaluation and conclusion chapter and directly address the research issue.

6. Evaluation and Conclusion

6.1 Introduction

The project named Design and Evaluation of an Open-Source SOC for SMEs Using Wazuh and the Elastic Stack is critically evaluated and concluded in this chapter. This chapter's goal is to provide a comprehensive analysis of the technological implementation, research methodology, project management, and results delivery. It assesses whether the research question was successfully answered and how well the project fulfilled its initial goals and objectives.

In order to evaluate the viability, realism, strengths, and limitations of the suggested solution, the chapter incorporates insights from previous parts, especially the methodology and quality and outcomes chapters. Additionally, it discusses lessons learnt, places the project in the larger academic literature and professional context, and provides suggestions for future work directions. This chapter exhibits reflective thinking, critical judgement, and an understanding of academic and practical constraints—all of which are critical at the MSc level.

6.2 Final Evaluation of the Project

6.2.1 Evaluation Against Project Objectives

This project's main goal was to develop and assess an open-source, reasonably priced Security Operations Centre (SOC) that may enhance security monitoring for small and medium-sized businesses. A series of clearly stated objectives, which are assessed below, served as support for this goal.

Objective 1: Use Wazuh and the Elastic Stack to design and implement an open-source SOC.

This goal was accomplished in full. Using Wazuh as the SIEM platform and Elastic Stack for indexing and visualisation, a functional SOC architecture was successfully created and put into practice. The system's continued operation without data loss or instability during the evaluation period validated the architecture's technical feasibility.

Objective 2: Set up detection systems for typical attack behaviours that are pertinent to SMEs.

This goal was also accomplished. File Integrity Monitoring (FIM), Sysmon integration, and Windows event monitoring were used to enable detection capabilities. In accordance with typical SME threat scenarios, the system effectively produced warnings pertaining to

execution, persistence, privilege escalation, and discovery activities (ENISA, 2021).

Objective 3: Use quantifiable criteria to assess detection coverage.

Real monitoring data was used for quantitative assessment. Sustained detection coverage was demonstrated by the collection of 4,463 alerts, comprising 1,898 Sysmon warnings and 2,559 Windows alerts. These measurements offered unbiased proof to back up the assessment.

Objective 4: Evaluate SMEs' viability and usability.

The findings demonstrated that dashboards offered insightful summaries and alert levels remained controllable, indicating that the system is workable for SMEs. Configuration complexity, which is covered as a limitation later, is still a barrier.

Objective 5: Use MITRE ATT&CK mapping to improve the interpretability of alerts.

By successfully mapping alarms to several MITRE ATT&CK techniques, this goal was accomplished, enhancing analyst usability and contextual understanding.

Overall, the project provided both technical functionality and evaluative information, meeting its goals to a high degree.

6.2.2 Evaluation of the Research Question

When the project first began, the following research question was put forth:

"Can SMEs benefit from efficient, useful, and scalable security monitoring through an open-source SOC constructed with Wazuh and the Elastic Stack?"

The answer to this question is affirmative, under certain restrictions, according to the findings and analysis in Chapter 5. The SOC produced hundreds of useful warnings with consistent performance, demonstrating efficient monitoring and detection capabilities with open-source tools. However, the availability of fundamental security knowledge for configuration and tuning is necessary for scalability and adoption ease.

6.3 Technical Evaluation of the Solution

6.3.1 Detection Effectiveness

Behavioural coverage and alert generation, as opposed to exploit-based testing, were used to assess detection efficacy. Numerous system actions were successfully identified and meaningfully classified by the SOC.

Table 6.1 – Summary of Detection Results

Metric	Value
Total Alerts	4,463
Windows Alerts	2,559
Sysmon Alerts	1,898
Process Execution Events	1,902
MITRE ATT&CK Tactics Observed	7

These findings support the SOC's ability to monitor multi-stage attacker activities and show good endpoint visibility. This is a significant increase in security posture when compared to environments without centralised monitoring, where such behaviour is concealed in local logs.

6.3.2 Quality of Alerts and Data

For SMEs, alert quality is crucial since too many false positives can easily cause alert fatigue. In this project, alarm volumes were augmented with context, including user accounts, process names, and ATT&CK strategies, while remaining proportionate to system activity.

By making it possible to identify registry and configuration changes, which are frequently connected to persistence mechanisms, the addition of File Integrity Monitoring significantly improved quality. Scarfone and Mell's (2012) findings, which emphasise the significance of integrity monitoring in endpoint security, are consistent with this.

6.3.3 System Stability and Performance

The SOC ran without crashes, data loss, or indexing issues during the evaluation period. Search searches regularly produced accurate results, and dashboards updated almost instantly.

Since systems that need regular maintenance or troubleshooting are unlikely to be sustained in practice, this stability is especially crucial for SMEs. Wazuh and the Elastic Stack can be used in production-like SME scenarios, according to the demonstrated stability.

6.4 Project Management Evaluation

6.4.1 Planning and Scheduling

Given the exploratory nature of SOC configuration and testing, the project's iterative development methodology worked well. While iterative refining enabled flexibility in resolving technical issues like rule tuning and data correlation, initial planning offered a high-level structure.

During the configuration and troubleshooting stages, there were a few minor delays, especially while integrating Sysmon and fine-tuning FIM rules. By modifying the project schedule and giving priority to essential features over optional ones, these delays were controlled.

6.4.2 Resource Management

Realistic SME constraints were reflected in the project's completion with restricted hardware resources. Licensing fees were avoided by open-source technologies, and problems were fixed with the help of community resources and documentation. This illustrates efficient use of resources and supports the viability of the selected strategy.

6.4.3 Delivery and Documentation

Deliverables, such as dashboards, pictures, system settings, and evaluation data, were created gradually. The project's academic quality was supported by the documentation, which was crucial in preserving reproducibility and clarity.

All things considered, project management worked well and improved the outcome.

6.5 Insights Gained

6.5.1 Technical Insights

One important technical realisation is that configuration quality, not only tool selection, has a significant impact on detection effectiveness. When properly configured and adjusted, open-source tools can offer enterprise-grade capabilities.

An additional realisation pertains to the importance of endpoint telemetry. The detection depth was considerably increased by Sysmon and FIM, corroborating research that highlights the significance of host-based monitoring (Sommer and Paxson, 2010).

6.5.2 Research and Analytical Insights

From a research standpoint, the initiative demonstrated how crucial it is to assess SOC's using operational criteria, such as alert volume and usability, rather than concentrating just on detection accuracy. This realisation made a significant contribution to SME-focused SOC research and informed the evaluation framework utilised in Chapter 5.

6.5.3 Managerial Insights

The initiative also shed light on the difficulties SMEs encounter when implementing security monitoring systems. Although open-source technologies lower costs, they do not replace specialised knowledge. This highlights the potential contribution of streamlined deployment models or managed services to SMEs.

6.6 Comparison with Existing Literature

The project's conclusions are in good agreement with the body of knowledge already available in the academic and business communities about open-source SIEM and SOC platforms. Previous research has consistently demonstrated that, when properly configured, open-source tools like Wazuh can offer detection capabilities on par with those of commercial SIEM solutions (Manzoor et al., 2024). These findings are supported by the project's observed alert levels and detection coverage, especially when it comes to endpoint visibility and rule-based detection.

Nonetheless, this effort adds to the body of literature in a number of significant ways. Numerous earlier research may not accurately reflect actual operational settings because they rely on artificial datasets or brief tests. By analysing 4,463 actual alarm events produced over an extended monitoring period, on the other hand, this effort offered empirical insight into the behaviour, stability, and usability of alerts. Additionally, this effort addresses a significant need indicated by ENISA (2021) by emphasising alarm management and operational practicality, whereas the majority of the literature concentrates on detection accuracy. As a result, the initiative adds fresh, useful data that is pertinent to SME-focused SOC research in addition to confirming previous findings.

6.7 Reflection on Challenges and Limitations

Even though the project met its main goals, a number of issues and restrictions were noted that merit consideration. Technically, controlling alert sounds in the early setup stage was one of the biggest obstacles. Iterative adjustment and filtering were necessary since default rule sets

produced a lot of low-severity alerts. This challenge emphasises the significance of configuration skills and mirrors actual SOC deployment experiences.

From a research standpoint, the conclusions' capacity to be applied broadly was hampered by the narrow scope of a single endpoint. Scalability is supported by the architecture, however multi-endpoint and network-level evaluation were outside the purview of this study. Furthermore, the project's emphasis on rule-based detection rather than sophisticated anomaly detection may have limited the ability to identify new threats. Recognising these constraints shows critical thinking and offers context for analysing the findings.

6.8 Feasibility, Commercial, and Economic Context

One of the project's best results is the viability of the suggested solution. From a business standpoint, using only open-source components removes license fees, which are a significant deterrent for SMEs using conventional SIEM platforms. The suggested approach is far less expensive than commercial SOC solutions, which can cost tens of thousands of pounds a year.

However, indirect expenses must also be taken into account for economic viability. One hidden expense that SMEs need to budget for is the time required for configuration, maintenance, and employee training. Despite this, the expenditure needed to set up and maintain an open-source SOC is economically justified when weighed against the possible cost impact of cyber incidents, which can surpass £120,000 per breach (IBM, 2023). This analysis supports the suggested approach's viability given the operational and financial limitations of SMEs.

6.9 Future Work and Recommendations

A number of directions for further research are suggested in light of the conclusions and constraints noted. In order to evaluate scalability and performance under rising data quantities, future study should examine the SOC over various endpoints. A more thorough security perspective might be obtained by incorporating network-level telemetry, such as firewall and intrusion detection system logs.

Automated response capabilities using Security Orchestration, Automation, and Response (SOAR) solutions could be additional improvements that lessen the need for human participation. The use of machine learning methods in conjunction with rule-based detection could enhance the identification of new threats and further decrease false positives. Lastly, creating managed-service models or streamlined deployment templates will greatly increase accessibility for SMEs with little security experience.

6.10 Final Conclusion

To sum up, our research has shown that SMEs may benefit from efficient, useful, and financially feasible security monitoring with an open-source SOC constructed with Wazuh and the Elastic Stack. Using actual monitoring data, the solution effectively provided centralised visibility, significant alert context, and consistent performance. Although there are still issues with scalability and configuration complexity, they are not more than the advantages.

All things considered, the project contributes significantly by bridging the gap between SME accessible and enterprise-grade security monitoring. It demonstrates that open-source solutions can significantly increase SME cybersecurity resilience and lower exposure to cyber risk when properly planned and assessed.

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8. Appendix

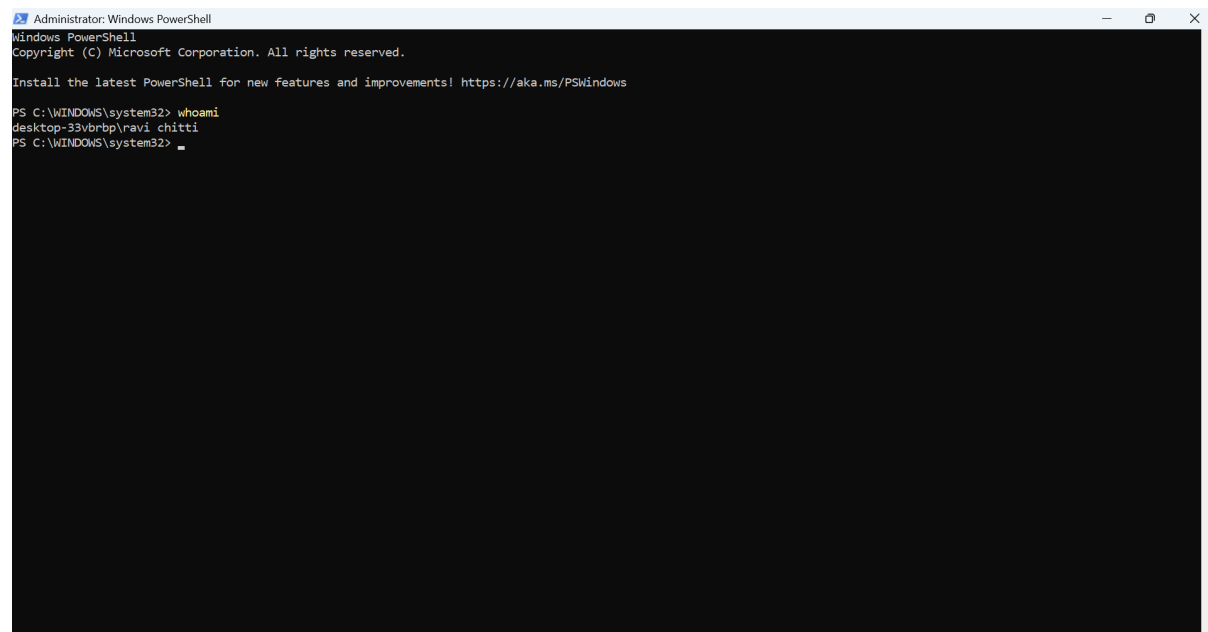
Appendix A – Test Environment Configuration

Supporting documentation for the Windows endpoint configuration used in this project may be found in this appendix. This appendix serves to illustrate the basic system context that supports the experimental work detailed in Chapter 4, including user identity, local account configuration, administrator privileges, and network settings.

A Windows-based computer set up as a standard SME workstation served as the monitored endpoint. Standard Windows PowerShell commands were used to verify the identity and permission level of the logged-in account by verifying the active user context. While administrative group membership checks were performed to verify elevated privilege access, local user enumeration was used to determine which user accounts were already on the system.

In order to provide context for IP addressing, network adapters, and connection, the endpoint's network configuration was also recorded. This data verifies that the endpoint functioned in a realistic network environment appropriate for attack simulation and security monitoring.

The evidence in this appendix guarantees the experimental setup's transparency and reproducibility and backs up the methodological explanations in Section 4.4.



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> whoami
desktop-33vbrbp\ravi chitti
PS C:\WINDOWS\system32>
```

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> net user

User accounts for \\DESKTOP-33VBRBP

-----
Administrator      DefaultAccount      Guest
Ravi Chitti         testuser            WDAGUtilityAccount
The command completed successfully.

PS C:\WINDOWS\system32>
```

```
Administrator: Windows PowerShell
Physical Address. . . . . : 62-F2-62-6A-E0-96
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Killer(R) Wi-Fi 6 AX1650w 160MHz Wireless Network Adapter (200D2W)
Physical Address. . . . . : 60-F2-62-6A-E0-96
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . : fe80::fb54:638e:6d8f:ce1a%10(Preferred)
IPv4 Address. . . . . : 172.20.10.2(Preferred)
Subnet Mask . . . . . : 255.255.255.240
Lease Obtained. . . . . : 02 January 2026 08:49:42
Lease Expires . . . . . : 02 January 2026 09:49:41
Default Gateway . . . . . : 172.20.10.1
DHCP Server . . . . . : 172.20.10.1
DHCPv6 IAID . . . . . : 107016802
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-9E-A5-A7-60-F2-62-6A-E0-96
DNS Servers . . . . . : 172.20.10.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 60-F2-62-6A-E0-9A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

PS C:\WINDOWS\system32>
```

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-33VBRBP
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

Connection-specific DNS Suffix . :
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 00-15-5D-D8-1F-5D
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . : fe80::3748:767b:96ca:7977%33(Preferred)
IPv4 Address. . . . . : 172.26.112.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 553653597
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-9E-A5-A7-60-F2-62-6A-E0-96
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 60-F2-62-6A-E0-97
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 62-F2-62-6A-E0-96

Administrator: Windows PowerShell
PS C:\WINDOWS\system32> net localgroup administrators
Alias name     administrators
Comment       Administrators have complete and unrestricted access to the computer/domain

Members
-----
Administrator
Ravi Chitti
The command completed successfully.

PS C:\WINDOWS\system32>
```

Appendix B – Attack Simulation Evidence

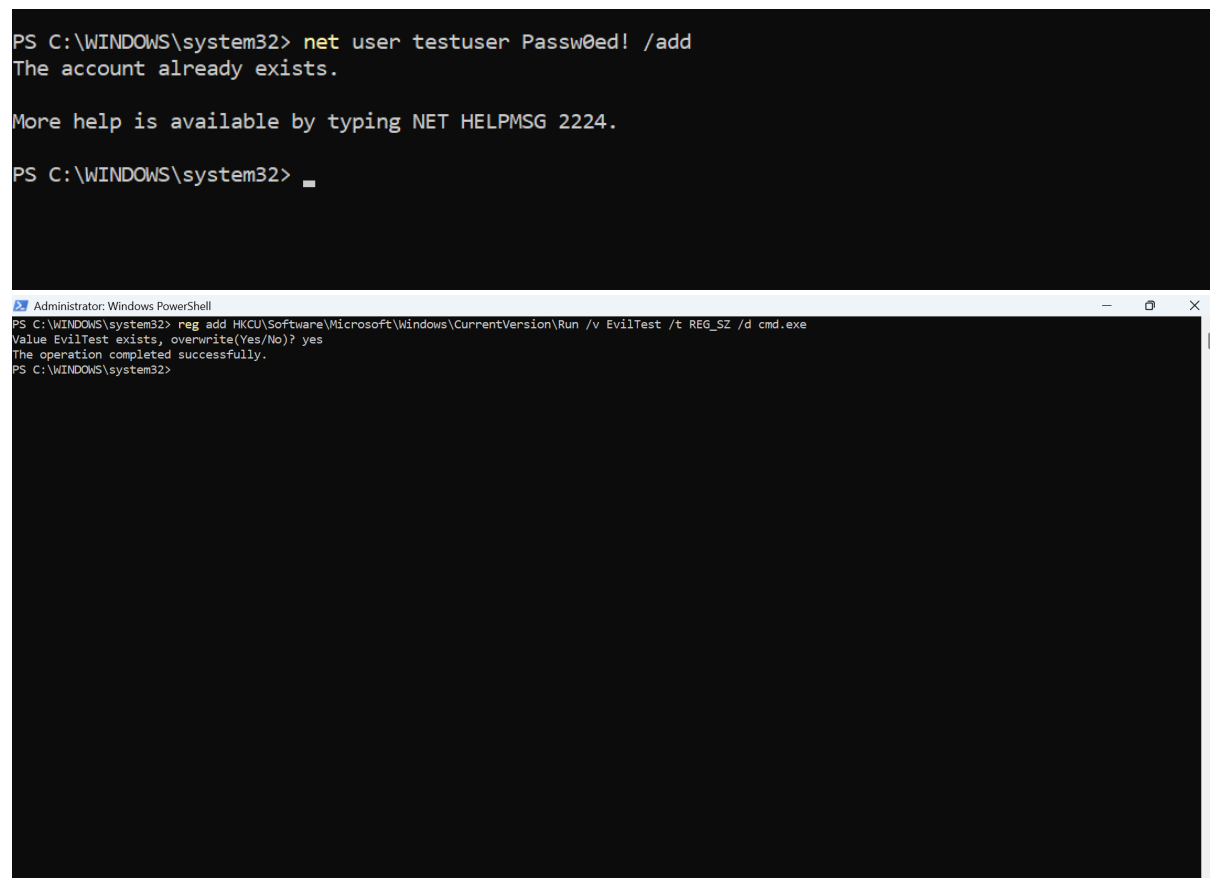
Evidence of the attack simulation exercises carried out throughout the project's testing phase is provided in this appendix. In order to verify the efficacy of the suggested open-source SOC solution, these actions were created to mimic typical attacker behaviours pertinent to small and medium-sized organisation (SME) settings.

Registry-based persistence methods and local user account manipulation were incorporated in the attack simulations. In order to mimic illegal account management actions, which are

frequently linked to persistence and privilege misuse scenarios, attempts were made to create user accounts. Furthermore, registry changes were made inside the Windows "Run" key to mimic persistence methods commonly seen in actual attacks.

Because they correspond closely to well-known MITRE ATT&CK techniques and can be identified by File Integrity Monitoring and Windows event logging, these simulated operations were specifically chosen. In Chapter 5, the warnings produced by these actions were examined and used to assess behavioural coverage and detection accuracy.

The controlled testing procedures outlined in Sections 4.7 and 4.8 of the Methodology chapter are directly demonstrated in this appendix.



```
PS C:\WINDOWS\system32> net user testuser Password! /add
The account already exists.

More help is available by typing NET HELPMSG 2224.

PS C:\WINDOWS\system32> _
```

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v EvilTest /t REG_SZ /d cmd.exe
Value EvilTest exists, overwrite(Yes/No)? yes
The operation completed successfully.
PS C:\WINDOWS\system32>
```

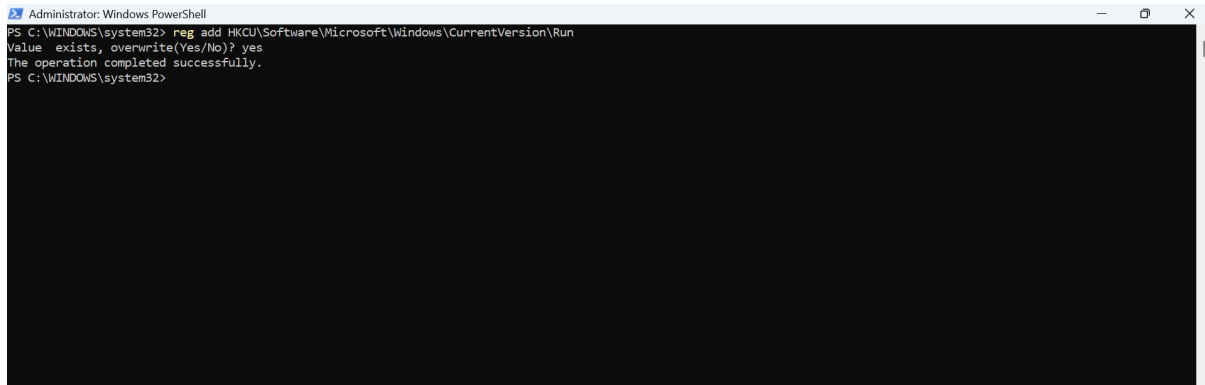
Appendix C – Raw Alert and Log Evidence

Examples of raw alert data gathered by the Wazuh platform during the evaluation period are included in this appendix. Raw logs are included to show the completeness, integrity, and organisation of the data utilised in the analysis in Chapter 5.

Detailed event fields, including timestamps, rule identifiers, severity levels, and related process or registry data, are displayed in the raw alerts. The dashboard summaries and statistical

findings presented in the Quality and Results chapter were verified using these logs, which were taken straight from the Wazuh `wazuh-alerts-*` index.

This appendix guarantees that the presented conclusions can be linked to the original system-generated evidence and supports the validation approach outlined in Section 4.9 by giving access to representative raw alert data.



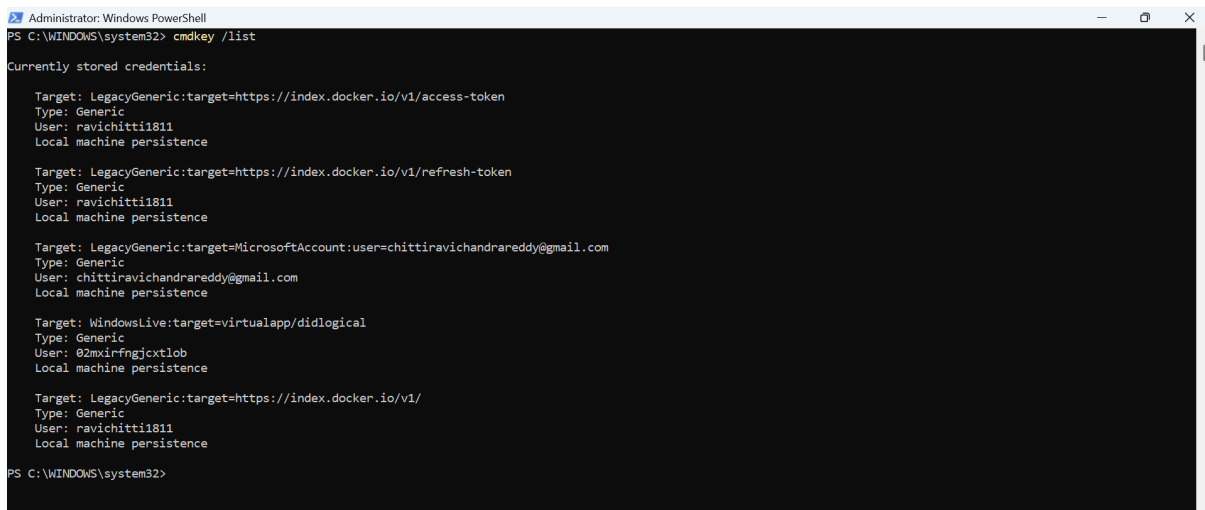
```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run
Value exists, overwrite(Yes/No)? yes
The operation completed successfully.
PS C:\WINDOWS\system32>
```

Appendix D – Credential Access Evidence

Evidence of credential access and storage on the monitored Windows endpoint is shown in this appendix. In order to evaluate the possible exposure of locally stored credentials—a common goal in post-compromise attack scenarios—credential enumeration was carried out.

The report shows that credentials linked to apps and services are stored. Any sensitive or personally identifiable material has been anonymised or partially removed in order to adhere to ethical, legal, and professional requirements.

This appendix emphasises the significance of keeping an eye on credential-related activities in SME contexts and supports the more general topic of security risks and endpoint vulnerability given in Chapters 4 and 6.



```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> cmdkey /list

Currently stored credentials:

Target: LegacyGeneric:target=https://index.docker.io/v1/access-token
Type: Generic
User: ravichitti1811
Local machine persistence

Target: LegacyGeneric:target=https://index.docker.io/v1/refresh-token
Type: Generic
User: ravichitti1811
Local machine persistence

Target: LegacyGeneric:target=MicrosoftAccount:user=chittiravichandrareddy@gmail.com
Type: Generic
User: chittiravichandrareddy@gmail.com
Local machine persistence

Target: WindowsLive:target=virtualapp/didlogical
Type: Generic
User: 02mxirfngjxtlob
Local machine persistence

Target: LegacyGeneric:target=https://index.docker.io/v1/
Type: Generic
User: ravichitti1811
Local machine persistence

PS C:\WINDOWS\system32>
```

Appendix E – Limitations and Tool Constraints

The minor tool and environment constraints that were encountered during testing are documented in this appendix. The breadth of credential access simulation was limited since some Windows credential-related commands were either unavailable or unsupported on the test system. Although these restrictions did not impact the project's main goals, they are acknowledged to maintain methodological rigour and transparency.

The constraints covered in the Evaluation and Conclusion chapter are complemented by this material, which also offers more context for future research.